Trade Tracker — System Diagrams

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Figure 4.1: Three-Tier Architecture Model

Three-Tier Architecture — Trade Tracker (MERN Stack)

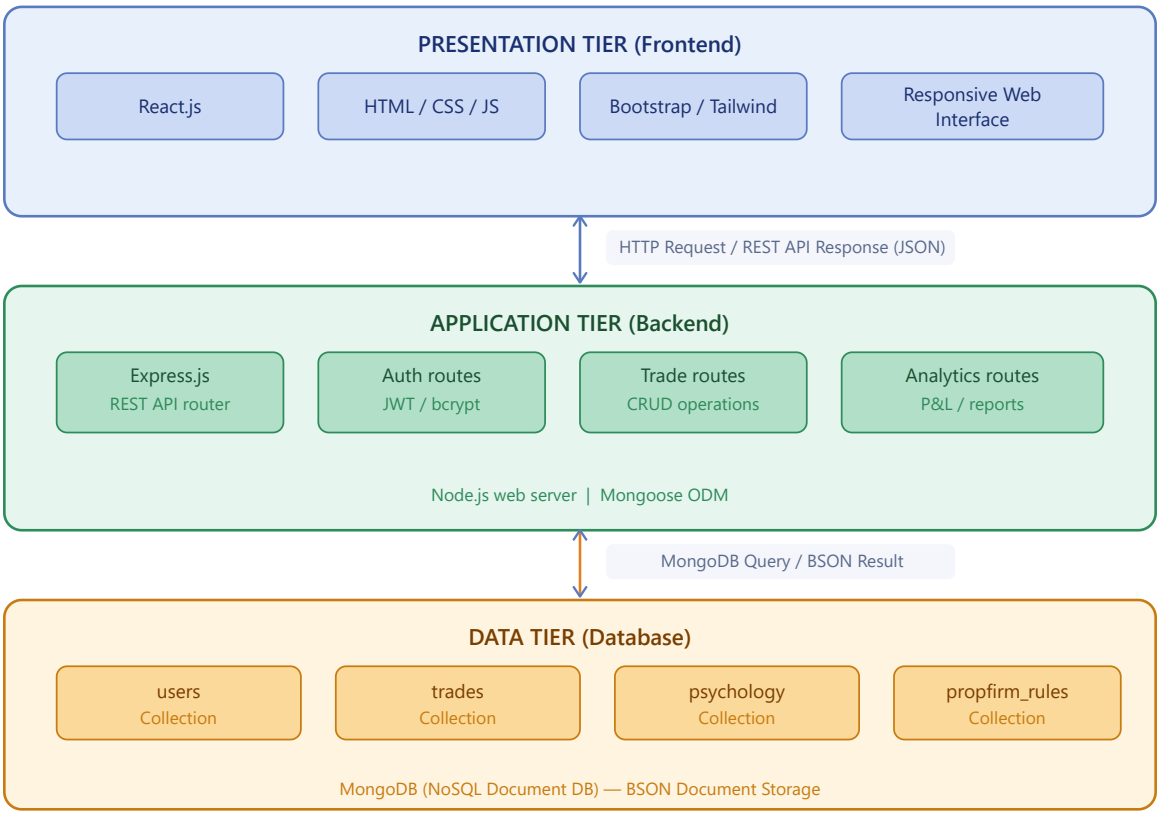


Figure 4.2: Context-Level DFD (DFD-0)

High-level data flows between external entities and the Trade Tracker System

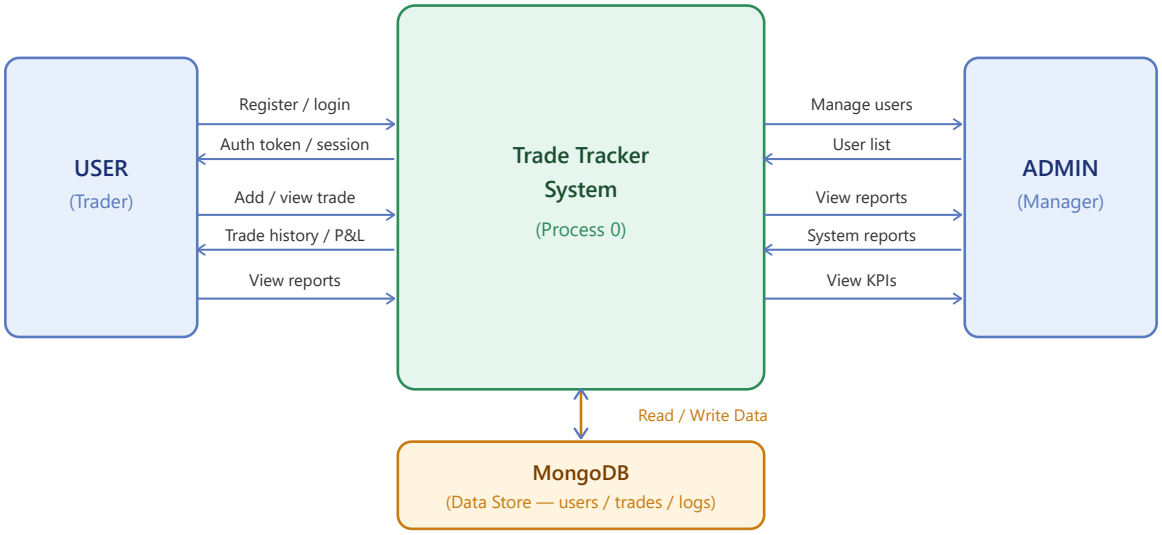


Figure 4.3: Level-1 DFD (DFD-1)

Main system processes and data store interactions

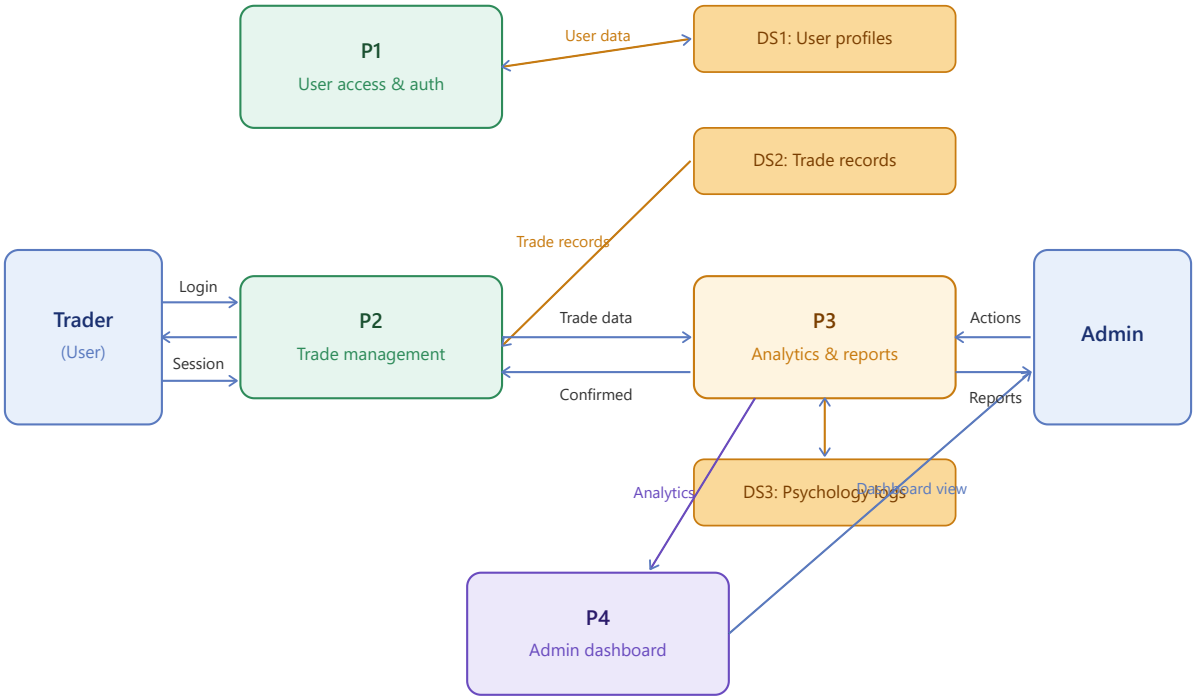


Figure 4.4: Level-2 DFD — Add Trade Process

Detailed workflow of recording a new trade entry

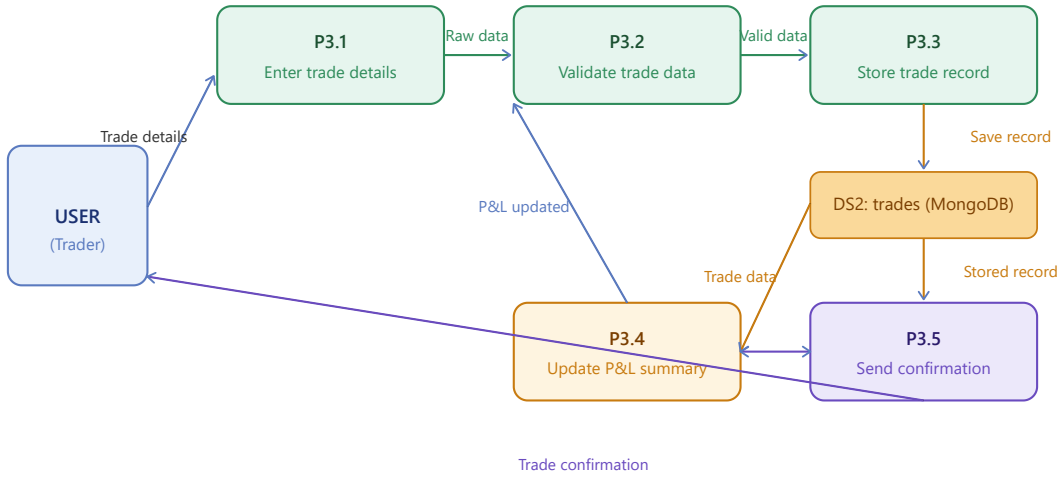


Figure 4.5: Entity-Relationship (E-R) Diagram

MongoDB collections and their relationships for Trade Tracker

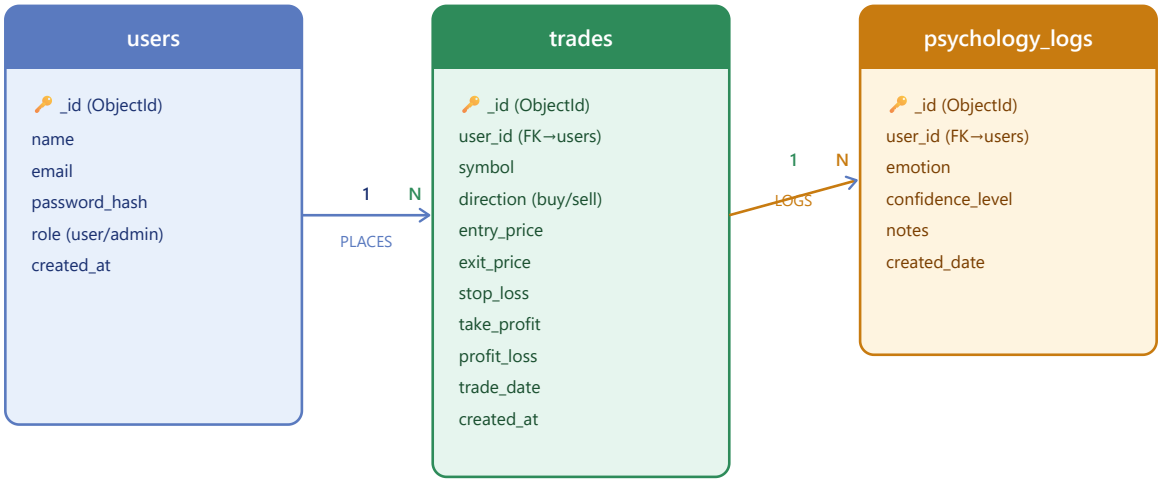


Figure 5.1: Use Case Diagram

Interactions between actors (User / Admin) and Trade Tracker system use cases

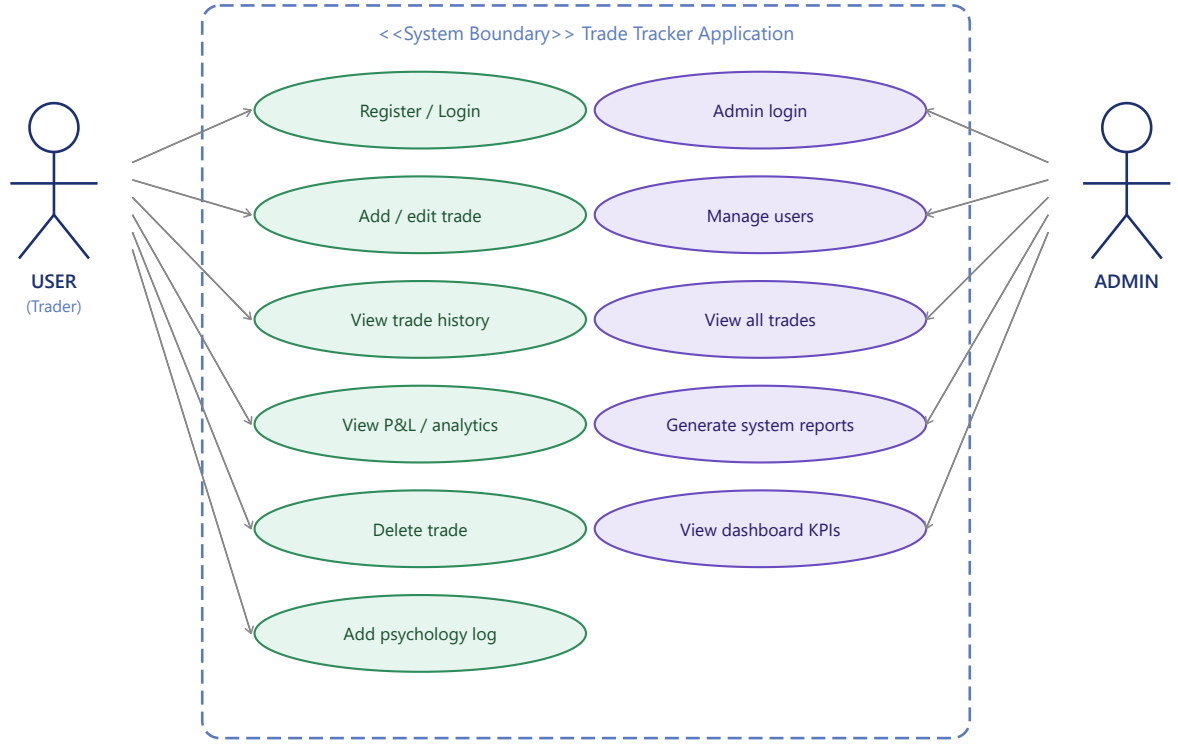


Figure 5.2: Activity Diagram — User Registration

Step-by-step flow for creating a new trader account

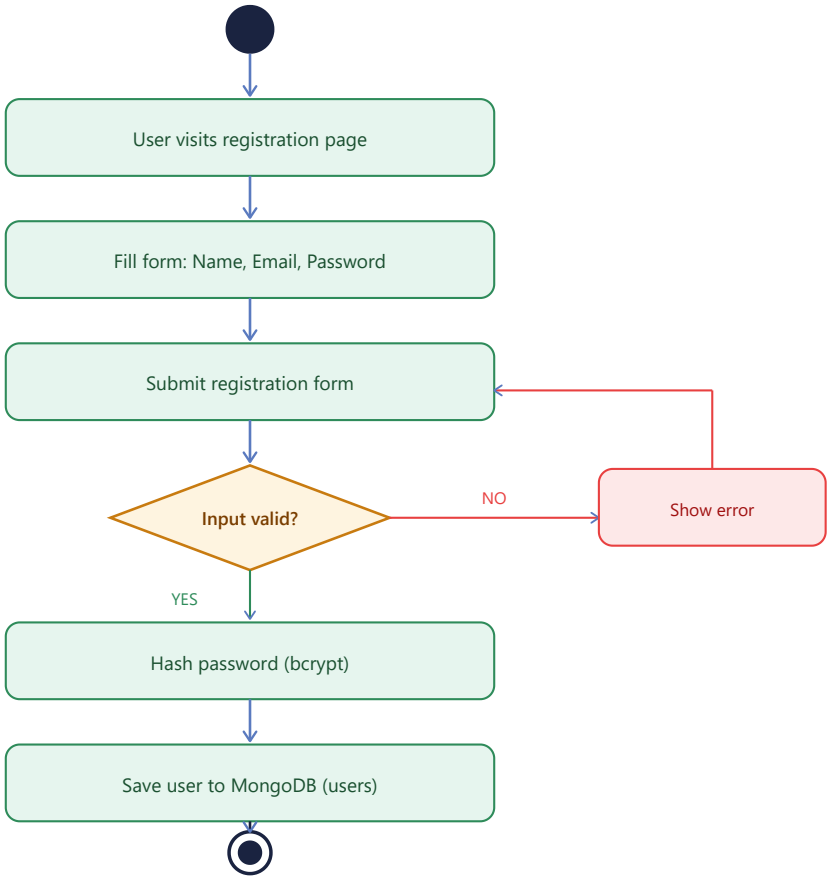
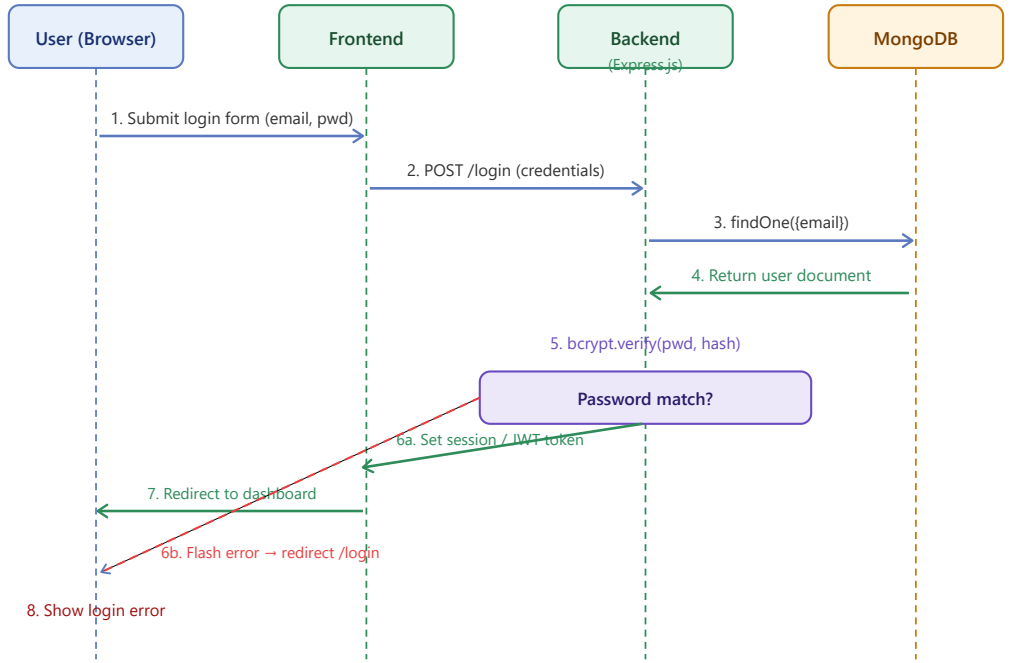


Figure 5.4: Sequence Diagram — User Login

Interaction between components during the authentication process


User (Browser)
Frontend
Backend
(Express.js)
MongoDB
1. Submit login form (email, pwd)
2. POST /login (credentials)
3. findOne({email})
4. Return user document
5. bcrypt.verify(pwd, hash)
Password match?
6a. Set session / JWT token
7. Redirect to dashboard
6b. Flash error → redirect /login
8. Show login error